



after two-and-a-half months?”

Imagine working on a cake for two-and-a-half months and then finding out that the customer is not happy with the cake. You will either have to start the whole process again or leave it for a new cake. Considering the same in the world of semiconductors, Industry 4.0, on the other hand, has the capability of signal corrections that are needed to be made at each stage during these two-and-a-half months. This opportunity Industry 4.0 provides can help dramatically improve die yield.

But isn't implementing Industry 4.0 expensive for the already expensive semiconductor industry?

Investment is indeed a big and core topic for everyone. The best way to predict the prospects of the investment is to start with the subscription services. That way you can easily set short-term goals, track results, and determine the ROI, which will guide and justify additional investments. As a matter of fact, at Advantest we advise our smaller customers to start with the subscription model first. This way it becomes part of the operating expense (OPEX) for a company.

I think the best way to find out if something is expensive or not is by calculating the amount of investment in terms of return on investment (ROI). There's always a pay-back period involved. It varies from company to company but typically, if the ROI stretches over decades, you can conclude that it isn't a good investment.

Would the subscription services require the provider to go to the client locations physically?

One of the key benefits of Industry 4.0 lies in providing services virtually. A lot of the things can be done virtually. At the end of the day, it is the clients who know their machines best. Advantest encourages

customers to come up with their own set of applications, and the required training would definitely be looked into.

The semiconductor market is highly competitive, and survival clearly goes to the fittest. Our data shows that companies that are technologically ahead are also ahead in time to market, when compared to their competitors. Though, there is no standard on the adoption of Industry 4.0, it all depends on the maturity of the company as well. Some companies are lagging and have not adopted even the Industry 3.0 features, which places them at a competitive disadvantage.

The decision of adopting Industry 4.0 must come from the executive level. The ROIs are not related to one business aspect but affect multiple businesses throughout the entire company. It is good to know that almost all leading companies in the semiconductor industry have taken initiative in Industry 4.0 domain and this was done at the executive level.

Would implementation of Industry 4.0 in the semiconductor industry result in shrinkage of jobs?

That fear, in my opinion, is very overrated. Every country and every company is trying to improve its better living index. For every job that is being replaced there are multiple jobs being created by the same industry. The biggest example of this is the number of data scientists required in the industry. The designation data scientist did not even exist a few years ago.

Industry 4.0 generates big data and at Advantest we are still in the process of gathering data in terms of gigabytes per die. Much of this data was not being looked at in the past, and there can be instances where important data is not even collected. Industry 4.0, in terms of the semiconductor industry, is creating more jobs. **EFY**

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